Project Proposal

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	15
Clarity	3					
Content Quality	6					
Spelling & Grammar	3					
Organization and Formatting	3					
					Total obtained mark	
Comments						

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BlindGuide

1. Introduction

Visually impaired people often face difficulties while moving independently and understanding their surrounding environment. Traditional walking aids can help detect physical obstacles but cannot identify objects, read text, or understand complex situations. **AI Vision Assistant** is a mobile-based intelligent system that uses a camera and Artificial Intelligence to understand the user's surroundings. The system provides real-time voice guidance to help visually impaired people move more safely and independently.

1.1 Background Overview

Visually impaired individuals may struggle with identifying obstacles, recognizing objects, reading signs, and navigating unfamiliar places. Existing assistive tools often provide limited information and may require additional hardware. Recent advancements in Computer Vision, OCR, AI, and Speech Technology make it possible to develop affordable mobile-based assistance systems. This project aims to combine these technologies into a single AI-powered assistant that can understand the environment and provide useful information through voice.

1.2 Problem Statement

Visually impaired people cannot easily identify objects, obstacles, text, or environmental conditions around them. Traditional walking sticks mainly detect physical obstacles but cannot provide contextual information about the surroundings. This creates difficulties when navigating unfamiliar environments, reading signs, or identifying important objects. Therefore, an intelligent system is needed that can see, understand, and describe the environment through voice**.**

1.3 Objectives

General Objective

The general objectives of the project are to:

- Develop a mobile-based assistive system for blind and visually impaired people.
- Help users navigate their surroundings more safely and independently.
- Provide real-time environmental information through voice guidance.
- Assist users in identifying obstacles, objects, and important environmental information.
- Provide basic navigation and emergency assistance.
- Reduce the dependency of visually impaired people on others during daily activities.

AI Objectives

The AI-specific objectives of the project are to:

- Develop AI-based real-time object detection for identifying surrounding objects.
- Use Computer Vision to detect obstacles and potential hazards.
- Apply AI-based scene understanding to interpret the user's surroundings.
- Use OCR and AI to recognize and read text from signs, labels, and documents.
- Implement AI-based proximity estimation to identify nearby obstacles.
- Generate context-aware voice instructions based on the detected environment.
- Use AI to prioritize critical situations and provide timely safety warnings.

1.4 Scope

In-Scope

The proposed system shall contain:

1. AI Object Detection

- The smartphone camera will capture the surrounding environment.
- AI will detect common objects such as people, vehicles, chairs, doors, stairs, bicycles, and other obstacles.
- The detected objects will be identified in real time.

2. Obstacle & Proximity Detection

- The system will identify objects that may block the user's path.
- AI will estimate whether an object is near, at medium distance, or far.
- The system will generate an immediate voice warning when a potentially dangerous obstacle is detected nearby.

3. AI Scene Understanding

- The system will analyze multiple objects within a scene.
- AI will determine which information is important to the user.
- The system will generate a simple description of the surrounding environment.

4. AI Text Recognition

- The system will capture visible text using the smartphone camera.
- OCR will recognize text from signs, room numbers, labels, notices, and documents.

- The recognized text will be converted into speech.

5. Basic Navigation Assistance

- The system may use GPS to provide basic outdoor navigation assistance.
- The user can provide a destination.
- The system will provide basic directional instructions through voice.

6. Emergency Alert

- The system may identify potentially dangerous situations.
- The user can trigger an emergency alert through voice or the application.
- Emergency contacts may receive an alert and location information.

Out-of-Scope

The system will not:

- Completely replace a human guide or professional mobility assistance.
- Guarantee 100% accurate object or hazard detection.
- Make autonomous decisions that could directly put the user at risk.
- Provide medical diagnosis or medical advice.
- Guarantee safe road crossing or fully autonomous navigation.
- Identify every person or object with perfect accuracy.
- Work as a fully autonomous vehicle or navigation system.

1.5 Stakeholders

Blind and Visually Impaired Users:
They are the primary users of the system and will use the application for environmental understanding, obstacle detection, text reading, and navigation assistance.

Family Members/Caregivers:
They may receive emergency notifications and location information when the user requires assistance.

Rehabilitation and Accessibility Organizations:
They may use the system as an assistive technology to support visually impaired individuals.

System Administrator:
Responsible for system configuration, user management, security, and system monitoring.

Development Team:
Responsible for designing, developing, training, integrating, testing, and maintaining the AI and software components.

1.6 Proposed Solution

AI Vision Assistant will provide an AI-powered real-time assistive system that works primarily through a smartphone camera and audio output. The system will continuously capture visual information from the user's surroundings and analyze it using AI-based object detection, obstacle detection, proximity estimation, scene understanding, and text recognition.

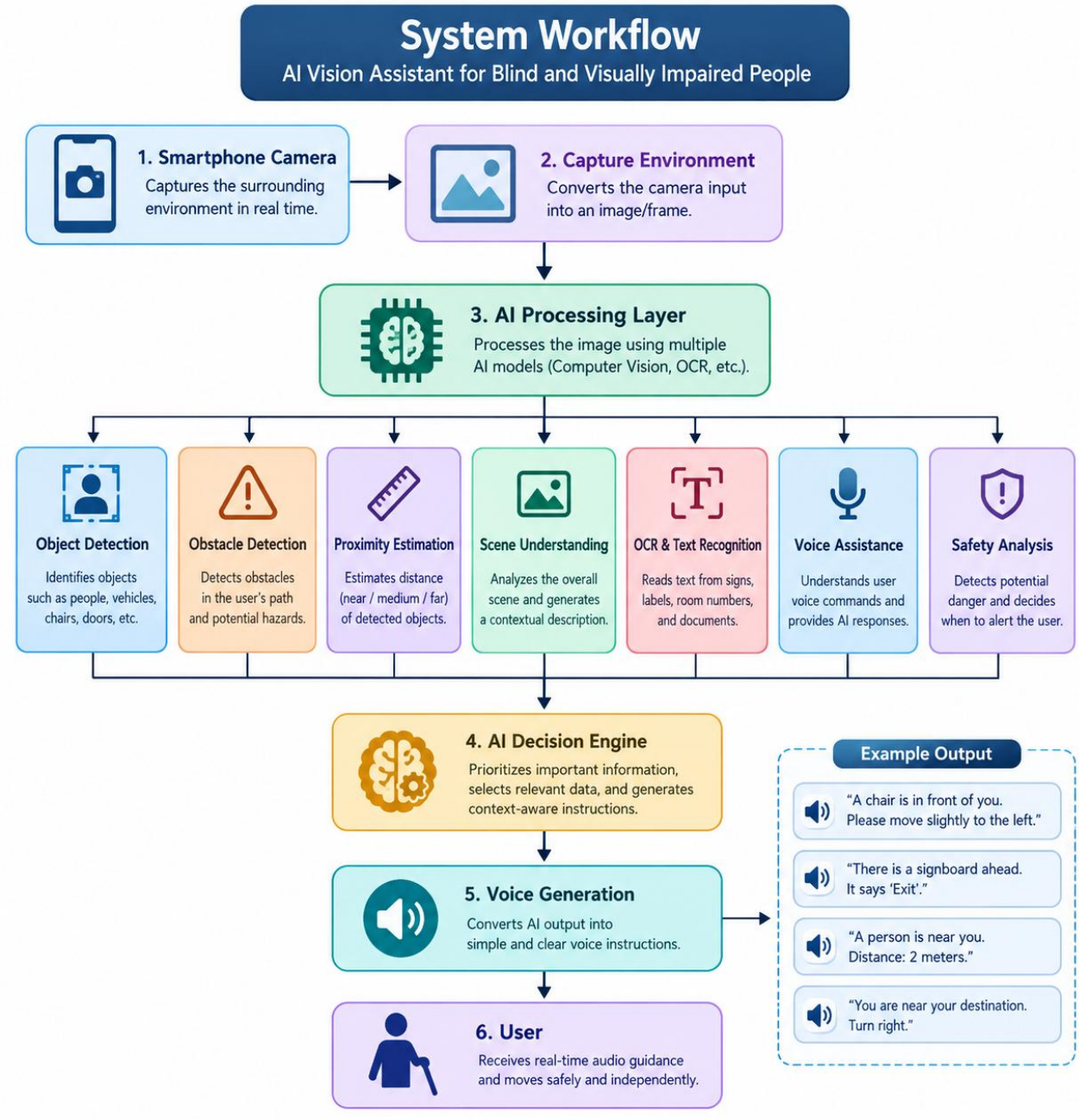
The system will identify important objects and potential obstacles around the user and determine which information is relevant to the user's safety. For example, if a chair or person is detected directly in the user's path, the system can provide a voice warning such as:

“There is an obstacle in front of you. Please move slightly to the left.”

If a signboard or room number is detected, the OCR module will recognize the text and read it aloud to the user.

The system will also support voice-based interaction, basic GPS navigation, and emergency assistance. Therefore, instead of simply recording the environment like a normal camera, AI Vision Assistant will analyze, understand, and communicate the environment to the user in real time.

System Workflow



1.7 AI Features

Proposed AI Features

1. Object Detection

Input: Smartphone camera live data
Output: Detected objects with approximate location information.

2. Obstacle Detection

Input: Detected objects + user's movement/path
Output: Safe / Potential Obstacle / Critical Obstacle.

3. Proximity Estimation

Input: Detected object and visual information
Output: Near / Medium / Far proximity status.

4. Scene Understanding

Input: Multiple objects detected in the current scene
Output: Contextual description of the surrounding environment.

5. OCR & Text Recognition

Input: Camera image containing visible text
Output: Extracted text and voice-readable information.

6. AI Voice Assistance

Input: User's voice command + camera/environment data
Output: Context-aware voice response.

7. AI Safety Analysis

Input: Objects, proximity, scene context, and environmental information
Output: Safety warning / Normal situation.

AI Techniques

The project may use:

- Computer Vision
- Object Detection
- Scene Understanding
- Image Processing
- OCR
- Distance/Proximity Estimation
- Machine Learning
- Speech Recognition
- Text-to-Speech
- AI/Vision-Language Models

Why AI Is Essential

AI forms the foundation of AI Vision Assistant. Without AI, the system would simply be a normal camera and audio application that cannot intelligently understand the user's surroundings.

AI is required to:

- Identify objects from camera images.
- Distinguish obstacles from normal environmental objects.
- Estimate the importance and proximity of detected objects.
- Understand the context of multiple objects in a scene.
- Recognize and interpret visible text.
- Determine which information is important for the user.
- Generate context-aware safety warnings.
- Provide intelligent voice assistance.

For example, a normal camera can capture an image containing a person, chair, road, and vehicle, but it cannot independently determine which object is dangerous or what information should be communicated to a visually impaired user. AI can analyze the visual information, understand the context, prioritize important events, and generate an appropriate voice response.

2. System Requirements

Functional Requirements

ID	Requirements	Description
FR-01	User Registration & Login	The system shall allow users to register and securely log in to the application.
FR-02	User Profile Management	The system shall allow users to manage their basic profile and accessibility preferences.
FR-03	Camera Integration	The system shall access the smartphone camera and capture the surrounding environment in real time.
FR-04	AI Object Detection	The system shall use AI to detect and identify common objects such as people, vehicles, chairs, doors, stairs, bicycles, and other relevant objects.

FR-05	Obstacle Detection	The system shall detect potential obstacles located in the user's walking path.
FR-06	Proximity Estimation	The system shall estimate the approximate distance of detected objects and classify them as near, medium, or far.
FR-07	AI Scene Understanding	The system shall analyze multiple detected objects and generate a meaningful description of the surrounding environment.
FR-08	OCR & Text Recognition	The system shall recognize and extract text from signboards, room numbers, labels, notices, and other visible documents.
FR-09	Text-to-Speech	The system shall convert recognized text and AI-generated information into understandable voice output.
FR-10	Voice-Based Assistance	The system shall accept basic voice commands or questions from the user and provide appropriate responses.
FR-11	AI Safety Analysis	The system shall analyze detected objects, obstacles, proximity, and scene information to identify potentially dangerous situations.
FR-12	Real-Time Voice Warning	The system shall provide immediate voice warnings when a potentially dangerous obstacle or situation is detected.
FR-13	Navigation Assistance	The system shall provide basic GPS-based navigation and directional voice guidance to a selected destination.
FR-14	Emergency Alert	The system shall allow the user to trigger an emergency alert and share their current location with configured emergency contacts.
FR-15	Safety Event History	The system shall record important detected events such as safety warnings, navigation events, and emergency alerts for authorized viewing.

Non-Functional Requirements

ID	Requirements	Description
NFR-01	Performance	The system shall process camera input with minimal delay to provide near real-time assistance.
NFR-02	AI Detection Accuracy	The AI models shall achieve an acceptable level of accuracy in object detection, obstacle detection, OCR, and scene understanding under normal operating conditions.
NFR-03	Response Time	The system shall generate voice warnings within a few seconds after detecting an important obstacle or safety event.
NFR-04	Availability	The system shall remain available during active assistance sessions and recover gracefully from temporary service failures.
NFR-05	Reliability	The system shall provide consistent assistance and minimize false warnings and missed critical situations.
NFR-06	Scalability	The system architecture shall support adding new AI models, object classes, accessibility features, and navigation capabilities without major architectural changes.
NFR-07	Security	The system shall protect user accounts, location information, camera data, emergency contacts, and other sensitive information from unauthorized access.
NFR-08	Privacy	The system shall handle camera images, recognized text, location data, and user information securely and only use them for intended assistance purposes.
NFR-09	Usability	The system shall provide a simple voice-based and accessible interface suitable for blind and visually impaired users.

NFR-10	Maintainability	The system shall use a modular architecture so that AI models, OCR, voice services, navigation, and emergency modules can be updated independently.
NFR-11	Compatibility	The system should support commonly available Android smartphones with compatible cameras, GPS, microphone, and speaker/headphone functionality.

AI Requirements

The system shall:

1. Detect and recognize common objects through smartphone camera data.
2. Identify potential obstacles in the user's walking path.
3. Estimate the approximate proximity of detected objects.
4. Understand the overall visual context of the surrounding environment.
5. Recognize and extract text using OCR.
6. Generate simple and meaningful descriptions of detected scenes.
7. Analyze environmental information and identify potentially dangerous situations.
8. Prioritize important information before generating voice instructions.
9. Convert AI-generated information into context-aware voice guidance.
10. Support intelligent interaction between the user and the AI assistant through voice commands.

3. Tools and Technologies

Programming Languages

- Python
- JavaScript

Frontend

- React Native / Flutter
- Android Studio
- Android SDK

Backend

- Node.js

- Express.js
- REST API

AI & Computer Vision

- Python
- OpenCV
- YOLO
- Machine Learning libraries
- Vision-Language / AI model
- Scene Understanding model
- Distance / Proximity Estimation techniques

OCR & Text Processing

- Tesseract OC32.0R / EasyOCR
- Image preprocessing
- Natural Language Processing where required

Database

- MySQL / PostgreSQL
- SQLite for local mobile data where required

Navigation & Location

- GPS
- Google Maps API / OpenStreetMap
- Location services
- Route and directional information

Communication

- SMS API
- Push Notification
- Emergency Contact / Calling Service

4. Project Timeline and Work Plan

The development of **AI Vision Assistant** will be completed through milestone-based phases over the entire project duration. The three team members will work on separate core modules while collaborating during AI integration, mobile application development, testing, and final documentation.

4.1 Milestone-Based Timeline

Milestone	Duration	Major Activities
Requirement Analysis & Research	Week 1-2	Problem analysis, literature review, requirements collection, accessibility research, and feature finalization
System Design	Week 3-4	System architecture, database design, UI/UX design, use-case diagram, workflow, and other system diagrams
AI Model & Dataset Development	Week 5-7	Dataset collection, preprocessing, object detection, obstacle detection, scene understanding, and AI model preparation
OCR & Voice Module Development	Weeks 5–8	OCR integration, text recognition, speech-to-text, text-to-speech, and voice command processing
Backend & Database Development	Week 6-8	Backend API, authentication, user profile management, event history, emergency contacts, and database implementation
Mobile Application Development	Week 6-9	Accessible mobile interface, camera interface, voice interaction, AI result display, and user settings
Camera & AI Integration	Week 8-10	Smartphone camera integration, live image processing, object detection, obstacle detection, proximity estimation, and AI response integration

Navigation & Emergency Module	Weeks 9–11	GPS integration, destination selection, directional voice guidance, emergency contact management, and emergency alert functionality
AI & System Integration	Week 10-11	Integration of AI models with mobile application, backend, OCR, voice assistant, navigation, and safety decision engine
Voice Alert & Assistance System	Week 11-12	Real-time voice warnings, contextual instructions, AI-generated descriptions, and user interaction testing
Testing & Evaluation	Week 13-14	Functional testing, AI accuracy testing, usability testing, performance testing, accessibility testing, security testing, and bug fixing
Final Deployment & Documentation	Week 15-16	Final integration, deployment, report preparation, presentation, demonstration, and final project evaluation

4.2 Team Task Distribution

Team Member	Assigned Role	Responsibilities
Mahin Al Hasan	AI & Computer Vision Developer	• Dataset preparation • Object detection • Obstacle detection • Scene understanding • Proximity estimation • AI model training and evaluation • AI-camera integration

MD Muhaiminur Rahman Maruf	Backend & Database Developer	• Backend API development • Database design and implementation • Authentication and user management • AI service/API integration • OCR and text-processing support • Emergency alert integration • System security
Farhana Jannat Saki	Frontend & Hardware Integration Developer	• Mobile application development • Accessible user interface • Camera interface integration • Voice interaction • Navigation interface • Real-time voice alert display • Emergency interface • Final system integration

5. Optional Sections

5.1 Existing System

Several existing assistive technologies provide features such as obstacle detection, OCR, navigation assistance, screen readers, and object recognition. However, these solutions may focus on individual assistive functions rather than combining real-time environmental understanding, obstacle detection, proximity estimation, scene description, text recognition, voice assistance, navigation, and emergency support into one integrated system.

Traditional walking sticks can help users detect physical obstacles but cannot identify what the obstacle is or explain the surrounding environment. Similarly, conventional camera applications can capture visual information but do not automatically interpret the environment and communicate the most important information to the user.

AI Vision Assistant aims to combine AI-based object detection, obstacle detection, proximity estimation, scene understanding, OCR, voice assistance, navigation, and emergency support to provide a more comprehensive assistive solution for blind and visually impaired people

5.2 System Architecture

The **AI Vision Assistant** system will follow a modular and layered architecture to integrate the smartphone camera, AI models, OCR, voice processing, navigation services, backend services, database, and emergency communication mechanisms.

The Input Layer will collect live visual information from the smartphone camera along with user voice commands and GPS/location data. The collected camera data will be transferred to the AI Processing Layer, where AI models will perform object detection, obstacle detection, proximity estimation, scene understanding, and OCR-based text recognition.

The processed results will then be passed to the AI Decision and Context Analysis Layer, which will combine multiple AI outputs to determine which information is most important to the user. For example, if a chair is detected close to the user's walking path, the system may classify it as a potential obstacle and prioritize a voice warning.

The Voice Assistance Layer will convert the processed AI information into simple and understandable voice instructions using Text-to-Speech technology. The user can also provide voice commands or questions, which will be processed by the AI assistant.

The Navigation Layer will use GPS and map services to provide basic directional guidance. The Emergency Layer will allow the user to trigger an emergency alert and share their location with configured emergency contacts.

Finally, the Mobile Application Layer will provide the main interface through which the user receives real-time audio guidance and interacts with the system. The Database Layer will store necessary user information, preferences, emergency contacts, and relevant event history.

5.3 Future Work

Future versions of **AI Vision Assistant** may include:

- Advanced AI-based indoor navigation.
- More accurate real-time depth and distance estimation.
- Advanced pedestrian and road-crossing assistance.
- Recognition of traffic signals and road signs.
- Recognition of currency notes and important documents.
- Advanced facial recognition for known individuals with appropriate privacy controls.

- Offline AI processing for improved privacy and availability.
- Wearable device integration such as smart glasses.
- Integration with smart canes and other assistive IoT devices.
- Improved multilingual voice assistance, including Bangla.
- Personalized AI assistance based on the user's mobility preferences.
- Predictive hazard analysis to identify potential risks before they become critical.
- Integration with emergency services for critical situations.

Conclusion

AI Vision Assistant is proposed as an AI-powered real-time assistive system designed to help blind and visually impaired people better understand and navigate their surroundings. By combining AI-based object detection, obstacle detection, proximity estimation, scene understanding, OCR, voice assistance, navigation, and emergency support, the system aims to provide useful environmental information and safety guidance beyond traditional assistive tools.

The project demonstrates how Artificial Intelligence and Computer Vision can be applied to solve a real-world accessibility problem. AI plays a central role in the system because it enables the application to interpret visual information, identify important objects, understand environmental context, recognize text, assess potential hazards, and generate meaningful voice instructions.

With further development, AI Vision Assistant can be extended with advanced indoor navigation, road and traffic assistance, wearable devices, smart canes, multilingual support, offline AI processing, and predictive hazard detection, making it a scalable and practical assistive technology for blind and visually impaired people.